Abb Muhaimeen

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EDUCATION

University of Dhaka — **Dhaka, Bangladesh**
B.S. in Software Engineering — *Expected Graduation: September 2026*

- *Concentrations:* Generative AI and Software Engineering
- *GPA:* 3.81/4.00
- *Related Coursework:* Artificial Intelligence, Software Requirements Specifications and Analysis, Distributed Systems

RESEARCH EXPERIENCE

UIUC++ Summer Research Program, University of Illinois Urbana-Champaign
Remote Research Intern | Advisors: Prof. Wing Lam, Prof. August Shi — *June 2026 – Present*

- Developing an LLM-based agentic tool that automatically implements Gherkin (Given/When/Then) step definitions and repairs Page Object Models, eliminating the manual coding step that Behavior-Driven Development workflows currently require of developers
- Designed an agent loop with tool support that explores the existing codebase (POMs, fixtures, XPaths), synthesizes the step-definition implementation, executes the resulting tests, and iterates on failures
- Integrated automated screenshot capture of executed scenarios so non-technical stakeholders can validate that a requirement was implemented correctly without a developer round-trip

University of Alabama
Remote Research Assistant | Advisor: Dr. Rayhanur Rahman — *May 2026 – Present*

- Investigating how small language models perform on malware classification using the EMBER dataset
- Running controlled experiments across model scales and prompting strategies to quantify accuracy-versus-cost trade-offs against conventional ML baselines

2025-Multimodal Parameter-Efficient Fine-Tuning Group
Remote Research Intern — *September 2025 – November 2025*

- Collaborated remotely with an international research group on a project focusing on PEFT techniques
- Executed PEFT methods like LoRa, QLoRa etc and tested its adaptability for multi-modality

IIT HCI Community
Research Intern — *September 2024 – January 2025*

- Contributed to ongoing research projects in Human-Computer Interaction (HCI) domain
- Assisted in conducting a literature review and Focus Group Discussions for an HCI-related paper

EXPERIENCE

Streams Tech Ltd
Research and Development Intern — *October 2025 – March 2026*

- Developed and deployed GenAI-powered modules for a production application, improving feature automation efficiency by 35% and reducing manual processing overhead
- Built practical Computer Vision pipelines (object detection, image analysis), increasing model accuracy by 12% and reducing inference latency by 25% through optimization and preprocessing

PROJECTS

Malware Copilot — *Advisor: Dr. Rayhanur Rahman*

- Built an AI-assisted malware analysis workbench that explains what a sample executable does rather than only classifying it, targeting the black-box behavior and false-positive burden of existing ML detectors
- Coupled Ghidra decompilation output with LLM reasoning to summarize and trace pseudocode behavior for analysts

- Implemented a RAG pipeline over a MITRE ATT&CK vector database that maps observed behavior to technique IDs with cited justifications, rendered on an ATT&CK Navigator-style heatmap
- Added function-level code embedding similarity search against known malware corpora to flag risky samples and surface the closest matching malware family

CharityGuard

- Engineered a decentralized Web3 charity platform using Node.js and Next.js, integrating blockchain to secure donation workflows
- Deployed on the Polygon Amoy Testnet using Hardhat, delivering an end-to-end working prototype

Connect Four

- Developed a Connect Four game in Lua, integrating the Minimax algorithm for intelligent move selection
- Enhanced AI performance by 25% through fine-tuning the evaluation heuristics function for faster decision-making

CubeMate

- Built a fully functional Rubik's Cube Solver in C++ during my freshman year, without relying on external libraries
- Implemented the complete algorithm from scratch, achieving 10% better performance than library functions

ACHIEVEMENTS

- **Problem Setter, Bangladesh AI Olympiad (BDAIO):** Authored competitive problems for the national AI olympiad; one problem was selected for the regional round
- **Codeforces Pupil** (Max Rating: 1324)

SKILLS

Programming: Python, Java, C/C++, SQL, Node.js, React.js, Lua, JavaScript/Typescript, HTML/CSS, Solidity

Tools: Git/Github, Linux/UNIX, MySQL, MongoDB, Figma, Jupyter Notebook, Google Colab, IntelliJ